

Cybersecurity Incident Report

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Project: Network Traffic Analysis - SYN Flood DoS

Course: Google Cybersecurity Professional Certificate

Supporting Evidence: TCP-HTTP-Log.pdf

Attack Assessment

This project analyzes a simulated incident using network logs provided through the Google Cybersecurity Professional Certificate. The assessment is based on the supplied TCP-HTTP log and does not represent an investigation of a live production environment.

The observed traffic is consistent with a TCP SYN flood denial-of-service (DoS) attack targeting the web server at 192.0.2.1 on TCP port 443. Legitimate clients initially complete TCP connections and receive successful HTTP responses. Repeated SYN packets from source IP 203.0.113.0 subsequently appear alongside connection resets affecting legitimate clients and a recorded HTTP 504 Gateway Timeout response.

Within the course scenario, a SYN flood is the most likely explanation for the service disruption. However, the supplied log does not independently confirm server resource exhaustion or identify the person responsible. The source IP is an observed traffic indicator rather than proof of an attacker's identity. The log also records an initial completed handshake from that source, so it would be inaccurate to state that every connection attempt remained incomplete.

The observed impact concerns service availability: legitimate users experience unsuccessful connections or delayed responses. The supplied evidence does not establish unauthorized data access, data modification, or data theft.

Evidence and Findings

The supporting evidence is contained in TCP-HTTP-Log.pdf, a course-provided simulated network log. The following observations support the attack assessment:

Normal traffic before the disruption: Rows 47–49 show client 198.51.100.23 completing the TCP three-way handshake with web server 192.0.2.1. Rows 50–51 record a request for /sales.html followed by an HTTP 200 OK response, demonstrating successful website access at that point.

Repeated SYN packets from one source: Rows 52–54 show an initial SYN, SYN-ACK, and ACK exchange involving source IP 203.0.113.0. Beginning at row 57, the same source repeatedly sends SYN packets to 192.0.2.1 on TCP port 443. This pattern continues throughout the remaining log.

Evidence and Findings Continued

Connection failures affecting legitimate clients: Rows 73, 80, and 85 show the server sending RST/ACK responses to clients 198.51.100.16, 198.51.100.7, and 198.51.100.22, respectively. These responses indicate that the connection attempts were reset. Their occurrence alongside the repeated SYN traffic supports the assessment of service disruption, but does not independently establish why the resets occurred.

Recorded gateway timeout: Row 77 contains an HTTP 504 Gateway Timeout response associated with client 198.51.100.5. This status indicates that a gateway or proxy did not receive a timely response from an upstream server. It provides additional evidence of an unsuccessful request, but does not independently prove that a SYN flood caused the failure.

Evidence limitations: The supplied log is a simplified educational dataset rather than a complete packet capture. It repeatedly uses the same source port for the suspicious SYN traffic, labels the HTTP response in row 77 as TCP, and repeats row number 214 near the end. It also contains an initial completed handshake from the suspicious source. These limitations prevent definitive conclusions about the number of half-open connections, server resource consumption, or the attacker's identity. The original log is preserved without alteration.

Taken together, the repeated SYN traffic, connection resets, and gateway timeout are consistent with the SYN flood DoS scenario. Confirmation in a production investigation would require additional evidence, such as a complete packet capture, server connection-state metrics, and server and gateway logs.

Technical Explanation

Normal TCP connection establishment

Before a client and server can exchange application data over TCP, they establish a connection through a three-way handshake:

1. **SYN:** The client sends a SYN packet to the server to request a connection.
2. **SYN-ACK:** The server acknowledges the request and sends its own synchronization information.
3. **ACK:** The client acknowledges the server's response, completing the handshake.

In this scenario, clients connect to the web server at 192.0.2.1 on TCP port 443. Rows 47–49 demonstrate a completed handshake, followed by a website request and a successful HTTP response in rows 50–51.

How a SYN flood disrupts service

A SYN flood exploits the server's handling of new TCP connection requests. When the server receives a SYN packet and responds with a SYN-ACK, it typically maintains a pending connection while waiting for the client's final ACK. This is called a half-open connection.

Technical Explanation Continued

An attacker can send numerous SYN packets without completing the corresponding handshakes. These pending connections can occupy the server's SYN backlog, which holds connection requests awaiting completion. If that capacity becomes exhausted, the server may be unable to accept additional legitimate connections until resources become available. The exact behavior depends on the server's configuration and protections.

Legitimate users may consequently experience connection delays, failed connection attempts, or timeouts. Repeated SYN traffic can also place additional processing demands on the server and network infrastructure.

Relationship to the observed evidence

The supplied log shows repeated SYN packets from 203.0.113.0 targeting 192.0.2.1:443, alongside resets affecting other clients and a recorded HTTP 504 Gateway Timeout response. These observations are consistent with the service disruption described in the course's SYN flood scenario.

However, the log does not directly show SYN backlog occupancy, server resource utilization, or the state of every connection. It also records an initial completed handshake from the suspicious source. Therefore, backlog exhaustion is a plausible technical explanation rather than a confirmed measurement from this dataset.

The HTTP 504 response represents an application-layer timeout: a gateway or proxy did not receive a timely upstream response. It is distinct from a TCP connection timeout. Although network disruption could contribute to that condition, the supplied evidence does not establish the gateway architecture or the exact cause of the upstream delay.